

CHAPTER 1

INTRODUCTION

The growing global demand for sustainable and environmentally friendly energy solutions has significantly accelerated the adoption of renewable energy technologies, particularly solar photovoltaic (PV) systems. Solar PV systems convert sunlight directly into electricity, offering a clean and inexhaustible energy source. However, a key challenge associated with solar PV systems is their intermittent and variable nature, primarily due to fluctuations in solar irradiance and weather conditions. This intermittency can lead to power quality issues, instability in the grid and inefficient energy utilization, particularly in standalone or weak-grid systems. To address these limitations, Battery Energy Storage Systems (BESS) are increasingly integrated with solar PV arrays. A BESS enables energy storage during periods of excess solar generation and provides power during periods of low or no solar availability, thereby enhancing system reliability, grid support and operational flexibility. In grid-connected configurations, such hybrid systems can perform critical functions such as peak shaving, load shifting, voltage and frequency regulation and energy arbitrage. This thesis focuses on the modeling and simulation of a grid-connected solar PV system integrated with a BESS, employing MATLAB/Simulink as the primary simulation platform. The objective is to develop a detailed dynamic model that captures the electrical behavior of the PV array, battery system, Direct Current- Direct Current (DC-DC) converters, inverter and grid interface. The model also incorporates advanced control strategies such as Maximum Power Point Tracking (MPPT), State of Charge (SOC) management, and grid-synchronization algorithms. Through time-domain simulation under various operating scenarios including variable irradiance and load profiles, this work evaluates the systems performance, energy flow and power quality characteristics. Aims to demonstrate the technical feasibility, performance advantages and practical implications of integrating BESS with grid-connected PV systems for modern power applications.

1.1 SYSTEM ARCHITECTURE

The modeled system consists of the following major components:

1. **Solar PV Array:** A Solar PV Array consists of interconnected PV modules that convert sunlight into electrical energy, forming the primary source of renewable DC

power. Its performance is influenced by environmental factors like irradiance and temperature, requiring precise modeling for optimal system integration.

2. **DC-DC Boost Converter:** A DC-DC Boost Converter is a power electronic device that steps up the input voltage from a solar PV array to a higher, regulated DC voltage level, enabling efficient energy transfer. It plays a crucial role in implementing Maximum Power Point Tracking (MPPT) to maximize power extraction from the PV system under varying environmental conditions.
3. **Battery Energy Storage System (BESS):** A Battery Energy Storage System (BESS) stores electrical energy chemically and delivers it on demand, enhancing grid stability and renewable integration by balancing supply and demand fluctuations. It comprises battery cells, power converters, and management systems to ensure safe, efficient, and reliable operation.
4. **Bidirectional DC-DC Converter:** A Bidirectional DC-DC Converter enables controlled power flow between the battery and the DC bus, facilitating both charging and discharging operations. It ensures efficient energy exchange in renewable systems by regulating voltage levels and maintaining battery health through precise current control.
5. **DC Link Capacitor:** A DC Link Capacitor stabilizes the DC bus voltage by smoothing voltage ripples and transient fluctuations between power conversion stages in renewable energy systems. It ensures reliable operation of inverters and converters by providing a steady DC voltage reference for efficient power transfer.
6. **DC-AC Inverter:** A DC-AC Inverter converts the DC power from solar PV arrays or batteries into grid-compliant AC electricity, enabling seamless integration with the utility grid or local loads. It employs advanced control techniques to regulate output voltage, frequency and power quality for efficient and stable operation.
7. **Grid Interface:** The Grid Interface facilitates the safe and efficient interconnection between the renewable energy system and the utility grid, managing power exchange and ensuring compliance with grid codes. It incorporates synchronization, protection, and control mechanisms to maintain stability and power quality during grid integration.

1.1.1 SOLAR PV ARRAY SYSTEM

A Solar Photovoltaic (PV) Array System is a renewable energy technology designed to convert solar radiation directly into electrical energy using the photovoltaic effect. It

plays a pivotal role in sustainable energy systems and is widely utilized in residential, commercial, and utility-scale applications. This section provides a comprehensive overview of the components, working principles, and design considerations of a PV array system.

1. Fundamentals of Photovoltaic Technology: The core of a PV system is the solar cell, a semiconductor device that converts sunlight into electricity. When solar irradiance strikes the surface of a PV cell, it excites electrons, creating electron-hole pairs. Through an internal electric field within the p-n junction of the cell, these charge carriers are driven toward the terminals, generating a direct current (DC).

2. CHARACTERISTICS OF A PV CELL

- **Open-circuit voltage (V_{oc}):** The Open-circuit voltage (V_{oc}) of a PV array is the maximum voltage available across its terminals when no load is connected, representing the upper voltage limit of the system. It depends on factors such as solar irradiance, cell temperature, and the number of series-connected cells, and is critical for designing system voltage ratings and safety margins
- **Short-circuit current (I_{sc}):** The Short-circuit current (I_{sc}) of a PV array is the maximum current flowing through the terminals when the output voltage is zero, occurring under a direct short-circuit condition. It is primarily influenced by solar irradiance and cell temperature, serving as a key parameter for array performance and protection design.
- **Maximum power point (P_{max}):** The Maximum Power Point (P_{max}) of a PV array represents the operating condition at which the product of current and voltage is maximized, yielding the highest possible power output. Tracking and maintaining operation at P_{max} is essential for optimizing energy harvest under varying environmental conditions.
- **Fill factor (FF):** The Fill Factor (FF) of a PV array quantifies the quality of the solar cell by expressing the ratio of the maximum obtainable power to the product of open-circuit voltage and short-circuit current. A higher FF indicates better cell efficiency and lower internal losses, making it a critical parameter for performance evaluation.
- **Efficiency (η):** The Efficiency (η) of a PV array system is the ratio of the electrical power output at the maximum power point to the incident solar power on the array surface, reflecting the system's capability to convert sunlight into usable

electricity. It is influenced by factors such as module materials, temperature, irradiance, and system losses, serving as a key metric for overall performance assessment.

1.1.2 PV MODULES AND ARRAYS

Since a single PV cell generates a relatively small voltage (typically $\sim 0.5\text{--}0.6\text{ V}$), multiple cells are electrically connected in series and/or parallel to form a PV module. Modules are further connected to form a PV array capable of meeting higher power demands.

- **Series connection:** In a series connection of a PV array, individual solar modules are connected end-to-end to increase the total output voltage, while the current remains equal to that of a single module, optimizing voltage for system requirements. It increases voltage.
- **Parallel connection:** In a parallel connection of a PV array, multiple solar modules are connected side-by-side to increase the total current output while maintaining the voltage equal to that of a single module, enhancing current capacity for load demands. It increases current.

A PV Array is a complete assembly of multiple PV modules interconnected to provide the required power output. The overall configuration is determined by the desired voltage, current and power output, as well as the characteristics of the load or power conversion system.

1.1.3 COMPONENTS OF A PV ARRAY SYSTEM

A typical Solar PV array system consists of the following key components:

Solar PV Modules

- It converts sunlight to DC electricity.
- It is made from silicon (monocrystalline, polycrystalline, or thin-film).

Mounting Structure

- It gives Mechanical supports the PV modules.
- It can be fixed-tilt or tracking (single-axis or dual-axis) to optimize solar exposure.

Maximum Power Point Tracker (MPPT)

- It ensures the PV array operates at its optimal power point regardless of varying solar irradiance and temperature conditions.

Inverter

- It converts the DC output of the PV array to AC for grid integration or AC loads.

Battery Energy Storage System (Optional)

- It stores excess energy for use during non-sunlight hours or grid outages.

Charge Controller

- It Basically regulates the voltage and current going to the battery to prevent overcharging or deep discharging.

Protection Devices

- It is basically used to protect devices, includes fuses, circuit breakers, surge protectors, and grounding equipment.

1.1.4 OPERATING PRINCIPLES

The electrical output of a PV array depends on several factors

- **Solar irradiance (G) [W/m^2]:** Solar irradiance is the measure of solar power received per unit area on the PV array surface, typically expressed in watts per square meter (W/m^2), directly influencing the electrical output and efficiency of the system.
- **Cell temperature (Tc) [$^{\circ}\text{C}$]:** Cell temperature in a PV array refers to the operating temperature of solar cells, which significantly affects their performance, efficiency, and output voltage, as higher temperatures typically reduce power generation capacity.
- **Incident angle and shading:** The incident angle is the angle at which sunlight strikes the PV array surface, affecting energy absorption, while shading reduces irradiance by blocking sunlight, both critically impacting the array's power output and efficiency.
- **Series and parallel configuration:** Series and parallel configurations of PV arrays combine modules to optimize voltage and current: series connections increase voltage by linking modules end-to-end, while parallel connections increase current by connecting modules side-by-side.

The current-voltage (I-V) and power-voltage (P-V) characteristics of a PV array are nonlinear and vary dynamically with environmental conditions. The maximum power point (MPPT) is the operating point at which the product of current and voltage ($I \times V$) is maximized. Modern PV systems use MPPT algorithms (e.g., Perturb and Observe, Incremental Conductance) to track the MPP in real-time.

1.1.5 DESIGN AND PERFORMANCE CONSIDERATIONS

Designing an efficient PV array system requires detailed assessment

- **Site solar potential (solar insolation and shading analysis):** Site solar potential determines the available solar irradiance and climatic conditions at a location, directly influencing the PV array's energy generation capacity, system design, and overall efficiency in renewable energy applications.
- **Orientation and tilt angle:** Orientation and tilt angle of a PV array optimize the incident solar irradiance by aligning the modules perpendicular to the sun's rays, maximizing energy capture and enhancing the system's overall power generation efficiency throughout the year.
- **Electrical matching with load or grid:** Electrical matching with load or grid ensures the PV array's voltage and current output align with the connected system's requirements, optimizing power transfer, minimizing losses, and maximizing energy generation efficiency and system stability.
- **Environmental conditions** (temperature, dust, humidity): Environmental conditions such as temperature, humidity, dust, and weather variability directly impact the PV array's performance by affecting solar irradiance, cell efficiency, and system degradation, thereby influencing overall energy generation and reliability.

Performance is typically evaluated using parameters like

- **Performance ratio (PR):** The Performance Ratio (PR) of a PV array quantifies the system's actual energy output relative to its theoretical potential, accounting for losses due to temperature, shading, and system inefficiencies, thus indicating overall energy generation effectiveness.
- **Capacity utilization factor (CUF):** The Capacity Utilization Factor (CUF) of a PV array measures the actual energy generated over a period relative to its maximum possible output, reflecting system performance, utilization efficiency, and the impact of environmental and operational conditions.
- **Energy yield (kWh/kWp):** Energy yield (kWh/kWp) of a PV array represents the actual electrical energy produced per kilowatt peak of installed capacity, indicating system productivity and effectiveness under specific environmental and operational conditions.
- **System losses:** System losses in a PV array including thermal, mismatch and soiling losses reduce energy generation by lowering cell efficiency causing power inconsistencies between modules and obstructing sunlight absorption on module surfaces.

1.2 BATTERY ENERGY STORAGE SYSTEM (BESS)

A Battery Energy Storage System (BESS) is an advanced electrochemical system designed to store electrical energy and release it when required. BESS plays a critical role in modern power systems, especially when integrated with intermittent renewable energy sources such as solar photovoltaic (PV) and wind. By storing surplus energy during periods of low demand or high generation and supplying it during peak demand or generation shortfall, BESS significantly enhances system reliability, stability and flexibility.

1.2.1 COMPONENTS AND ARCHITECTURE

A typical BESS consists of the following main components:

1. **Battery Pack:** Comprising multiple cells connected in series and parallel to form modules and racks, the battery pack is the core energy storage unit. Common chemistries include:
 - **Lithium-ion (Li-ion):** Lithium-ion (Li-ion) batteries in PV systems provide efficient, high-energy-density energy storage, enabling reliable power supply during low solar generation and enhancing overall system stability, efficiency, and renewable energy utilization.
 - **Lead-acid:** Lead-acid batteries in PV systems offer cost-effective, mature energy storage solutions that support energy availability during low solar output, though they have lower energy density and shorter cycle life compared to advanced battery technologies.
 - **Sodium-sulphur (NaS):** Sodium-sulphur (NaS) batteries in PV systems provide high-capacity, long-duration energy storage with excellent efficiency and cycle life, enabling effective energy management and reliable power supply for grid integration and renewable energy utilization.
 - **Flow batteries:** Flow batteries like vanadium redox systems offer scalable, long-duration energy storage with decoupled power and energy capacities, enhancing PV system reliability by efficiently storing excess solar energy and supporting grid stability during variable generation.
2. **Battery Management System (BMS):** The BMS monitors and controls battery parameters such as voltage, current, temperature, and State of Charge (SOC) to ensure safe and optimal operation. It also performs cell balancing and protects against overcharging, deep discharge, and thermal runaway.

3. **Bidirectional Power Converter (DC-DC or DC-AC):** This converter facilitates both charging (from DC bus or AC grid) and discharging (to load or grid) operations. It ensures voltage and current regulation and manages power flow direction.
4. **Energy Management System (EMS):** The EMS coordinates with the BMS and the power converter to determine optimal charge/discharge schedules based on load
5. **Depth of Discharge (DOD):** The percentage of battery capacity discharged relative to its maximum. High DOD reduces battery life in some chemistries.
6. **Round-trip Efficiency:** The ratio of energy retrieved from the battery to energy used for charging. It is typically 85-95% for Li-ion batteries.

1.2.2 MODELING CONSIDERATIONS

Battery modeling in simulation environments like MATLAB/Simulink typically uses:

1. **Electrical Equivalent Models:** Representing voltage dynamics using Thevenin or Rint models.
2. **Electrochemical Models:** Capturing internal reaction kinetics (used for research-level accuracy).
3. **SOC Estimation Algorithms:** Including Coulomb counting, Extended Kalman Filter (EKF), or neural network-based approaches.

1.2.3 FUNCTIONAL ROLE OF BESS IN PV SYSTEMS

Energy Storage and Dispatch

1. **Charge:** During periods of high solar generation (e.g., midday), excess energy is stored in the battery.
2. **Discharge:** During peak demand or low generation periods (e.g., evening or cloudy weather), stored energy is released to support the load.

Peak Shaving

1. Reduces peak power drawn from the grid by discharging stored energy during peak demand, thereby lowering demand charges.

Load Shifting

1. Enables consumption of stored solar energy during non-generation hours, improving self-consumption and system efficiency.

Grid Support Services

1. Provides ancillary services such as frequency regulation, voltage support, and reactive power compensation.

Backup Power Supply

1. Ensures uninterrupted power during grid outages, enhancing system resilience.

1.2.4 CONTROL AND ENERGY MANAGEMENT

The Energy Management System (EMS) ensures optimal control of energy flows:

1. **Priority-based operation:** Prioritize local load, then charge battery, then export to grid.
2. **SOC limits:** Maintain healthy operating range to enhance battery life.
3. **Forecasting:** Use solar irradiance and load predictions to plan charging/discharging.
4. **Grid-interactive logic:** Respond to real-time pricing, demand-response signals, or blackout events.

1.2.5 ADVANCED CONTROL STRATEGIES:

1. Model Predictive Control (MPC)
2. Fuzzy logic and AI-based optimization
3. Real-time SCADA monitoring

1.3 GRID-CONNECTED

Grid-connection plays a pivotal role in the operation, control, and optimization of solar photovoltaic (PV) systems integrated with battery energy storage systems (BESS). In a grid-connected solar PV system, the generated DC power from the PV array is converted to AC and synchronized with the utility grid to supply electricity in real-time. This interconnection enables the system to feed surplus energy into the grid during periods of high solar generation and draw power during low generation or high demand, ensuring a reliable and continuous power supply. The modeling and simulation of such systems necessitate accurate representation of the grid interface, including voltage and frequency synchronization, power flow management, and fault ride-through capabilities. The grid connection facilitates energy balancing by coordinating the charging and discharging of the BESS in response to grid requirements and PV generation variability. Moreover, it supports ancillary services such as voltage regulation, frequency stabilization and peak load shaving, enhancing grid stability and renewable energy penetration.

Therefore, incorporating grid-connection dynamics in the simulation framework is essential to analyze system performance, optimize control strategies and evaluate the impact of renewable integration on the power network. This holistic approach enables

effective design and operation of sustainable, resilient and economically viable solar PV-BESS systems.

CHAPTER 2

LITERATURE REVIEW

2.1 GENERAL

This thesis presents a comprehensive study on the modeling and simulation of a grid-connected solar photovoltaic (PV) array system integrated with a battery energy storage system (BESS). The integration aims to enhance energy reliability, ensure continuous power supply, and improve grid stability by balancing generation and demand. Advanced simulation tools are employed to analyze dynamic performance under varying environmental and load conditions, enabling optimization of system design, control strategies and operational efficiency. Sarina Adhikari and Fangxing Li (2014) shows a method for integrating and coordinating solar PV generator management with battery storage control and maximum power point tracking (MPPT) control to give support for an islanded microgrid in terms of voltage and frequency (V-f). Additionally, for the grid-connected mode, active and nonactive/reactive power (P-Q) control with solar PV, MPPT, and battery storage is suggested. The control algorithms demonstrate efficient coordination between energy storage charging and discharging control, MPPT control, and inverter V-f (or P-Q) control. While taking into account the scenario of fluctuating irradiance and battery state of charge (SOC) constraints, the article also demonstrates efficient coordination among participating micro resources. The outcomes unequivocally confirm that the suggested control measures are effective [1]. Kumar et al. (2015) presents a coordinated control technique for distributed generators incorporated into a DC microgrid (DCMG) in both islanded and grid-connected modes of operation is proposed. A variety of nontraditional energy sources, storage systems, and three-phase and single-phase ac loads are connected to the DCMG. In order to integrate the three-phase load and utility grid into the DCMG under different operating situations, a control strategy for a three-phase voltage source inverter has also been proposed. The feed-forward and feed-back control loops are combined in the suggested control approach. In two rotating direct- and quadrature-axis synchronous reference frames, two proportional integral controllers have been proposed for inner current control and ac voltage regulation, respectively, managing the positive and negative sequence components [2]. Han et. al (2020) presents a two-stage, multi-timescale resilient unit commitment and economic dispatch strategy to maximize microgrid performance. The initial phase combines real-time sub hourly and day-ahead

models, which implies that the day-ahead dispatch result needs to simultaneously meet the real-time criterion. Verifying the viability of the day-ahead dispatch result under the worst-case scenario while taking into account the high degree of uncertainty in renewable energy dispatch and load consumption is the second step. The suggested model combines solar PV units with battery energy storage systems (BESS) to create a hybrid solar-storage system. In order to maintain the hourly output of the combined solar-storage system constant, the BESS is crucial in balancing the varying output of solar PV modules [3]. Mir et. al (2024) expressed as solar photovoltaic parks (SPPs) are increasingly incorporated into power systems, thorough dynamic modeling is becoming crucial for security evaluation. Nevertheless, this type of modeling requires a lot of computing power. In order to reduce computational load while maintaining the dynamic behavior of detailed models, this research suggests a dynamic aggregate equivalent model of SPPs. A clustering method is developed based on these criteria, acknowledging that SPP dynamics during faults depend not only on control tactics but also on fault severity and irradiation levels. A representative dynamic aggregate model with a ramp recovery mechanism for active power is created for every cluster. The precision and effectiveness of the model are confirmed by means of comprehensive MATLAB/Simulink simulations. The OPAL-RT platform is used to further demonstrate real-time feasibility, proving the model's efficacy for dynamic security assessment [4]. Cekalovic et. al (2025) presents to optimize seawater pumping operations, this research investigates demand-side management (DSM) through the integration of solar energy and battery storage utilizing economic model predictive control (EMPC). It assesses maximum demand (MD) and real-time pricing (RTP) tariffs. For load shifting and peak demand management, the suggested concept makes use of batteries, photovoltaic (PV) systems, and reservoirs. Four situations are examined. Only reservoirs, reservoirs with PV, reservoirs with batteries, and reservoirs with both PV and batteries are the first four options. Large mining pump station simulations demonstrate that reservoir-based load shifting, which is improved by PV to reduce peak demand, can result in significant cost savings. Batteries by themselves provide very modest cost benefits, but PV and batteries together provide the greatest flexibility and savings. Adaptive, economical, and dependable operating under fluctuating energy prices and demand is made possible by EMPC-based DSM [5]. Sun et.al (2014) provide power to the grid for dispatch, this paper details research on a large-scale dispatchable grid-connected photovoltaic (PV) system rather than providing the power to a nearby load. The PV system uses a hybrid

electricity storage system that combines batteries and supercapacitors to maximize the value of the solar energy. Instead of concentrating on the grid-connected inverter, this research suggests a control technique that targets the DC power at the DC link. To illustrate the system's viability, two representative sets of actual data that were gathered from active websites are used. Lastly, the excellent performance and viability of the suggested system and control technique are illustrated using the simulation results [6].

Wang et. al (2018) shows a supercapacitor (SC)-based energy storage unit is used to improve the stability of a multimachine power system integrated with a large-scale hybrid wind-photovoltaic (PV) farm. A 300 MW aggregated wind turbine generator (WTG) and a 75 MW photovoltaic array make up the hybrid farm. Both are coupled to a single DC link using converters. A voltage-source inverter is used to transfer power from this link to the grid. Through a bidirectional DC/DC converter, the SC balances power variations caused by solar and wind variability. The SC is equipped with a PID-based supplementary damping controller (PID-SDC) to enhance the damping of low-frequency oscillations. The SC with PID-SDC successfully improves system stability and performance during disturbances, as shown by root locus and time-domain simulations conducted under various wind and solar conditions [7].

Khaled et al. (2024) Using sustainable resources, expresses the best hybrid microgrid design for Riphah International University (RIU), Lahore, guaranteeing a dependable and reasonably priced energy supply. A battery bank, a converter, solar photovoltaic (PV) panels, and the national grid are all integrated into the system. Energy availability, environmental impact, cost, campus load, solar irradiation, PV efficiency, battery life, grid energy costs, load shedding, emissions, and other important elements are taken into account. Using HOMER Pro software, simulations yielded 932 possible designs. With a net present cost (NPC) of \$0.483 million and a 99.3% usage of renewable energy, the chosen model was the most effective. On the other hand, the least successful plan, which used diesel and the grid, had an NPC of \$1.89 million and used no renewable resources [8].

Bairami et al. (2023) presents a unique MPPT technique that combines adaptive perturbation and observation with predictive current management, this work first offers a small-signal model of a photovoltaic (PV) system, DC-DC boost converter, and PWM generator. In order to direct the DC-DC converter's inductor current toward the reference, the technique forecasts future current values and uses current control to estimate PV impedance. To improve performance and lessen fluctuations close to the maximum power point, the cost function incorporates voltage and current ripple

minimization. Rapid variations in solar radiation are efficiently detected by the method, which then modifies the duty cycle appropriately. When compared to PID control, simulations on a 26.6 kW three-phase grid-connected PV system in MATLAB/Simulink demonstrate better performance, with THD reductions of 0.1% (current) and 0.16% (voltage) [9]. Boubakir et al. (2021) expresses a robust model-free controller for a grid-connected photovoltaic (PV) system comprising a PV generator linked to a three-phase grid via a DC/AC converter. The control aims are to extract maximum power from the PV source, manage reactive power exchange, and enhance the quality of current injected into the grid. Instead of relying on a detailed dynamic model, the controller uses an ultra-local model, which is continuously updated using a numerical differentiator based on the system's input-output behavior. The model-free control approach combines a classical feedback controller with a compensator to address internal parameter variations and external disturbances. Simulation results demonstrate the controller's effectiveness in ensuring stable operation, efficient power extraction, and improved current quality in grid-connected PV systems under varying conditions [10]. Jagadeesan et al. (2022) presents an active power regulation method with real-time maximum power point (MPP) estimation is presented in this work for a two-stage, single-phase grid-connected solar PV (SPV) system. Without utilizing energy storage, the suggested control technique provides frequency regulation by operating the PV panel at the MPP or on the right-hand side (RHS) of its characteristic curve. A PI controller is used to track the reference power to be extracted, which is determined by a reserve portion specified by the plant operator. While the DC-AC converter preserves DC-link voltage and unity power factor, the DC-DC converter regulates through switching. MATLAB/Simulink was used to model the system, which was tested on a 250 Wp panel with a 230 V, 50 Hz grid [11]. Strache et al. (2014) compares advanced inverter architectures for photovoltaic (PV) systems string inverters, power optimizers and microinverters across various locations and applications. It focuses on the impact of partial shading and weather conditions on system performance. To enhance energy output in shaded areas, a submodule-level converter topology is proposed and evaluated. A detailed MATLAB/Simulink cell-level model is used to simulate energy output over 30 years, considering cell mismatch and degradation. Simulation results highlight the strengths and weaknesses of each inverter type under different scenarios. The study finds that PV systems with fewer solar cells connected in series achieve higher energy yields, even in unshaded environments, due to reduced mismatch losses

and improved adaptability to variable conditions [12]. Ding et al. (2012) presents a MATLAB-Simulink-based model for photovoltaic (PV) modules with an S-Function builder for simulation and a regulated current source is presented in this study. The model takes into consideration how temperature, irradiance, and array configuration affect PV performance. Numerous array designs are examined under nonuniform irradiance, and the modeling results closely resemble the I-V and P-V curves found in outdoor experiments. The model does a good job of capturing the intricacy of output power dynamics, including the presence of many peaks under irregular circumstances. Additionally, it offers maximum power point tracking (MPPT) methods and electrical circuit simulation in the MATLAB-Simulink environment. The model is appropriate for examining general MPPT behavior since experimental validation validates its accuracy and excellent prediction abilities under both uniform and nonuniform irradiation situations [13]. Zhang et al. (2025) presents Adaptive Regulated Sparse Regression (ARSR), a unique statistical learning technique, is presented in this work for data-driven modeling and control of solar PV systems. In order to better capture system dynamics, ARSR introduces variable-specific sparsity control, in contrast to classic sparse regression, which uses a uniform hyperparameter across all state variables. This lowers computing complexity and increases accuracy, particularly when dealing with big lists of candidate functions. For both open-loop and closed-loop control, the ARSR approach successfully detects complicated dynamics in single-stage and two-stage PV systems. It also facilitates fault analysis, which expands its applicability beyond traditional techniques. Real-time simulations support the suggested method, demonstrating its efficacy in real-world PV control and diagnostics [14]. Li et al. (2024) presents voltage support in power networks with a large penetration of inverter-based solar PV systems, this research proposes an adaptive voltage control technique. By grouping PV inverters together, the method makes use of their idle capacity to regulate voltage at the point of common coupling (PCC). It does not necessitate precise system models, in contrast to conventional techniques. Rather, it dynamically linearizes the system by online identification and real-time input/output measurements. The inverter cluster is then managed by a data-driven cooperative adaptive controller. The suggested approach maintains voltage stability while adjusting to system changes. Numerical simulations in MATLAB/Simulink validate its practical usefulness for contemporary PV-integrated grids by demonstrating its efficacy and robustness [15]. Rahman et al. (2021) shows a nonlinear adaptive integral backstepping control strategy for a solar PV system's DC-

DC converter and a three-phase grid-connected inverter. Active power regulation, maximum power extraction, and system stability are all guaranteed by the controller. Additionally, it minimizes harmonic injection into the grid while providing stable performance across a range of meteorological conditions and outside disturbances. Negative-definite Lyapunov functions are used to provide stability, and an adaptive law improves the rejection of disturbances. To maximize PV power, the DC-DC converter is combined with the incremental conductance method. With the help of an LCL filter, the inverter regulates reactive power and provides the grid with clean active power. The correctness, resilience and improved performance of the scheme under real-world settings are confirmed by simulation and laboratory studies [16]. Stonier et al. (2018) expresses identify, diagnose, and manage faults in PV panels, batteries, semiconductor switches, and inverters, this study suggests an intelligent fault-tolerant system for solar photovoltaic (PV) inverters that uses a controller based on an artificial neural network. The solar PV and battery system are coupled to a cascaded multilayer inverter for DC-AC conversion. The capacity of the suggested design to continue power transmission even in the event of a malfunction or partial shading is one of its main advantages. Additionally, the study examines typical issues with PV energy conversion and offers suitable fixes. The system's capabilities are confirmed by testing on a 3 KWp solar PV system and simulations conducted with MATLAB 2016a/Simulink. The dependability and efficiency of the design are confirmed by the results, which show efficient power delivery and system functioning under problematic circumstances [17]. Lin et al. (2020) expressed integrated AC/DC grids, this research proposes a comprehensive nonlinear transient modeling approach for photovoltaic (PV) systems. It effectively simulates electromagnetic transients (EMTs) by leveraging GPU's parallel computing capabilities in addition to CPU. High-fidelity simulations are made possible by modeling each PV panel separately under various temperature conditions and nonuniform irradiation. Computational efficiency is increased by using a scalable PV array model with modifiable aggregation levels. The single-instruction multiple-thread design of the GPU allows up to 10 million PV modules to be simulated concurrently. The GPU is supplemented by CPU processing in low-parallelism settings. Furthermore, dynamic EMT co-simulation is optimized using a multi-rate technique that takes into consideration the different time-step tolerances of AC and DC grids. MATLAB/Simulink and DSA Tools/TSAT are used to validate the results, verifying their accuracy and effectiveness [18]. Yousef Mahmoud and Ehab F. El-Saadany (2017)

shows global peak mistracking and mismatch in series-connected modules, partially shaded PV systems frequently experience power losses. Reconfiguring PV modules can effectively lower these losses, but the accuracy of the current approaches is computationally demanding, which restricts their applicability in large-scale systems. Based on the greedy optimization concept, this work suggests a unique reconfiguration technique that effectively finds the ideal configuration with a notably shorter computing time. The suggested approach provides a more straightforward but efficient solution than conventional techniques that call for intricate dynamic programming. A case study on a large PV system and a variety of shading circumstances are used to illustrate its efficacy, revealing that it can reduce mismatch losses while enhancing the viability of real-time reconfiguration in shaded PV systems [19].

Udoka C. Nwaneto and Andrew M. Knight (2023) study examines the effective modeling of a single-phase, two-stage grid-connected photovoltaic (PV) system using the dynamic phasor (DP) approach. Two models-DP-Full, which offers a comprehensive depiction of the PV array, capacitor, boost converter, DC-link, and MPPT control-and DP-Simp, which streamlines the DC-side by combining the dynamics of the capacitor and inductor into a first-order function and calculating the MPP voltage and current analytically, are suggested. A commercial electromagnetic transient software is used to verify both models against a comprehensive switching model. The outcomes demonstrate that the DP models provide excellent accuracy at a notably shorter simulation duration. They are ideal for quick transient analysis in grids with substantial PV penetration because of their computational efficiency, particularly in multi-converter circumstances [20].

Radwan et al. (2024) presents stable operation under extremely weak grid circumstances and faults, this study suggests a grid-forming (GFM) voltage-source inverter (VSI) with direct current control for a hybrid wind-solar generator. Without the need for an intermediary DC-DC conversion, the GFM-VSI links directly to a wind turbine with a permanent magnet synchronous generator (PMSG) through a voltage-source rectifier (VSR) and a solar PV array, increasing efficiency. While the GFM-VSI harvests solar energy and provides grid voltage/reactive power, the VSR manages wind power extraction with reserve capabilities. A state-space model with tiny signals is created and examined. The system's stability and performance are confirmed by simulations and real-time experiments, which show that it performs better than grid-following control, which becomes unstable in similar weak grid conditions [21].

Kumar et al. (2017) expressed a single-input fuzzy-tuned PI controller and a multilayer fifth-order generalized integrator

(MFOGI) are used to suggest an adaptive control method for a single-phase, two-stage grid-connected partially shadowed solar PV system. Fast global maximum power point tracking (GMPPT) is guaranteed by the innovative Human Psychology Optimization (HPO) method. Under disturbances such as overvoltage, undervoltage, harmonics, frequency fluctuations, and DC offset, the MFOGI efficiently extracts the fundamental grid voltage component. During transients, the fuzzy-tuned PI controller dynamically modifies gains, and a PV feed-forward component improves response time. Laboratory tests under various insolation, temperature, and grid settings confirm the system, which is modeled in MATLAB. Outperforming traditional techniques and satisfying IEEE-519 requirements, the results demonstrate negligible current harmonic distortion and a much shorter GMPPT time [22]. Kacimi et al. (2023) study presents a hybrid maximum power point tracking (MPPT) control approach that combines Model Predictive Control (MPC), Particle Swarm Optimization (PSO) and Incremental Conductance (Inc Cond) for grid-connected solar systems. The maximum power point is precisely located by the hybrid Inc Cond-PSO method in a variety of environmental circumstances. The boost converter is then controlled by MPC, which tracks this point and maintains good performance even when parameters change. Furthermore, by reducing the difference between reference and expected power to choose inverter switching states, a two-level inverter under MPC control injects the generated power into the grid. Dynamic simulations in MATLAB/Simulink show how well the technique works to provide accurate MPPT with fewer power oscillations, improving system stability and efficiency overall [23]. Chen et al. (2016) shows Leakage current in photovoltaic (PV) systems, which is heavily reliant on parasitic capacitance between the PV panel and ground, is the subject of this research. This capacitance is typically approximated based on experience, which results in errors. To precisely compute panel parasitic capacitances, which are separated into cell-to-frame (CCF), cell-to-rack (CCR), and cell-to-ground (CCG) capacitances, a novel 2-D parasitic edge capacitance model is suggested. A pi-shaped circuit model is created based on this to forecast the PV array's leakage current. The accuracy of the suggested approach is confirmed by experimental measurements, MATLAB simulations and theoretical computations. This method improves safety and system performance by combining accuracy and simplicity to provide a useful and trustworthy instrument for assessing leakage current in PV systems [24]. Lupangu et al. (2020) expresses a model predictive control (MPC) optimum control technique for a photovoltaic (PV)/battery hybrid system is presented. By taking

into account solar insolation availability, battery state of charge (SOC), and variable electricity tariffs (VET), the strategy seeks to optimize earnings. To regulate load demand during times of low and high prices, a two-stage control algorithm makes use of VET, predicted solar insolation, and SOC. In the first step, MPC forecasts solar irradiance, load consumption, and energy price while predicting SOC. Additionally, it minimizes forecasting mistakes and compensates reactive power during scheduling. Through MATLAB/Simulink simulations, the control strategy is verified, showing enhanced performance under a range of conditions, including abrupt changes and parameter uncertainties, guaranteeing dependable and effective PV/battery system operation [25]. Singh et al. (2023) presents a fifteen-level packed U cell (FLPUC) converter for grid-connected solar photovoltaic (GCSPV) systems with eight active switches is presented. In comparison to traditional converters, this architecture uses a PV source and two floating capacitors to provide fifteen output voltage levels while using fewer components. The operation modes and switching states are described in depth in the study. Independent power injection control into the grid is made possible by active and reactive power (PQ) control. The perturb and observe (P&O) MPPT approach is used to guarantee the PV array's maximum power while maintaining unity power factor on the AC side. To maintain optimum output levels, PI controllers balance the voltages of floating capacitors. IEEE-519 power quality standards are met by the system's performance under various solar conditions, which is modeled in MATLAB/Simulink and verified by real-time emulation [26]. Wang et al. (2014) shows power smoothing approach for a 1 MW grid-connected solar photovoltaic (PV) power plant is suggested in this research. A vanadium-based hybrid energy storage system (HESS) A supercapacitor bank and a redox battery are employed to even out the PV plant's variable output power. By preventing the VRB from operating at low power levels and lowering the SCB's needed power rating to just one-fifth of the VRB rating, the HESS's power management system increases the VRB's overall efficiency. The MATLAB/Simulink and PLECS software environments have been used to model the PV plant including the HESS. Extensive simulation results validate the effectiveness of the suggested power control approach [27]. Zhang et al. (2021) showing simplified analytical models in the synchronous frame for large-signal simulation and small-signal analysis, this study tackles low-frequency oscillations seen in a solar photovoltaic (PV) farm. Two models are put forth that maintain grid-side converter controls and grid interconnection electromagnetic dynamics while ignoring PV dc-side dynamics. While

the second model depicts the PV dc side as a constant dc voltage behind an impedance, the first model assumes a constant input power from the PV's dc side. Time-domain simulations, admittance frequency responses, and eigenvalue stability analysis are used to validate these reduced models against a comprehensive electromagnetic transient (EMT) testbed. According to the results, the second model provides a useful tool for examining and comprehending PV farm oscillations by accurately predicting oscillation stability [28]. Sachidananda Sen and Vishal Kumar (2020) presents detailed models that assume continuous dc-bus voltage and ignore storage effects are addressed in this study by presenting a simplified small-signal model (SSM) for microgrids (MGs) combining renewable energy sources (RESs) such as solar PV and batteries. By incorporating the dynamics of the inverters' input dc and output ac terminals, the suggested reduced-order model improves computational efficiency while capturing generating intermittency. A low-voltage feeder and two PV sources with parallel RLC loads make up the MG test setup. Comprehensive time- and frequency-domain analyses in MATLAB simulations are used for validation, in addition to hardware-in-loop testing with a dSPACE processor and real-time digital simulator (RTDS). The model's correctness for both transient and steady-state operations is demonstrated by the results, offering a useful tool for effective MG system analysis [29]. Khazaei et al. (2020) presents Low inertia from renewable energy sources typically gives the system some time to be controlled. Inertia emulation for photovoltaic systems has just introduced. (PV) generation to increase energy efficiency and combat minor inertia. This article's primary goal is to examine the stability of a three-phase, two-stage PV system for smart grid applications that is based on virtual inertia. The modeling process takes into account the dynamics of the DC/DC and DC/AC converters, as well as the AC side filter, virtual inertia controller, and controllers of the DC/DC and DC/AC converters. The sensitivity of the design parameters on the stability of a grid-connected virtual inertia photovoltaic system is assessed using eigenvalue analysis. The results of the eigenvalue analysis are validated using time-domain simulations with the MATLAB/Simscape Power System toolbox [30]. Dubey et al. (2018) The design, modeling, and control of a quasi-resonant zero-current switching boost converter for a surface-mounted permanent magnet synchronous motor (SPMSM) driven by photovoltaic (PV) that drives compressor loads in distant refrigeration plants are presented in this study. Peak sun radiation is effectively matched with cooling demand by the solar-powered refrigeration system. A field-oriented controlled voltage source inverter that adjusts to changing solar irradiation is

used to manage the SPMSM drive's speed. MATLAB/Simulink is used to model and simulate the control system, which exhibits reliable performance under a variety of operating circumstances. In order to validate the simulation results and prove the system's efficacy for dependable, solar-based refrigeration in remote areas, an 800 W PV-powered SPMSM drive prototype is created and tested in the lab [31]. Ahmed Issa Alnahhal and Balazs Plesz (2024) shows performance of a combined photovoltaic (PV) and thermoelectric generator (TEG) system at low concentration levels is examined in this article in relation to solar spectrum splitting. The solar spectrum is separated into two parts: the within-bandgap spectrum (350–1100 nm), which is further split between the PV and TEG, and the post-bandgap spectrum (1100–4000 nm), which is focused towards the TEG. By altering the splitting wavelength in increments of 50 nm, fifteen combinations were examined. The findings indicate that while the TEG makes up for the PV's power loss, spectrum splitting lowers the temperature of the solar cell by lowering its thermal burden. The ideal splitting wavelength transfers more energy to the TEG at higher light concentrations, increasing overall efficiency. By lowering the temperature and raising the overall power production, the integrated system performs better than individual PV cells [32]. Su et al. (2014) discusses power quality issues such voltage fluctuations and unbalance in three-phase, four-wire low-voltage (LV) networks that are brought on by the growing usage of home photovoltaic (PV). It suggests a thorough approach to PV operational optimization that makes use of single-phase inverter actual power curtailment and latent reactive power to improve voltage magnitude, balance profiles, and lower generation costs and network losses. Using a weighted-sum approach, the multiobjective optimal power flow (OPF) problem is reduced to a single-objective problem, which is then resolved in MATLAB using global sequential quadratic programming (SQP). The plan is verified using smart meter data and thorough 24-hour simulations on an actual imbalanced LV network from the Perth Solar City trial in Australia. The advantages and real-world uses of the control technique are examined [33]. Shadab Murshid and Bhim Singh (2019) shows design and experimental verification of a fuzzy precompensated hybrid proportional-integral (PI) controller for a freestanding solar water pumping system powered by a permanent magnet synchronous motor (PMSM) are presented in this study. Due to their fixed gains, traditional PI controllers are sensitive to changes in parameters. The speed error is processed by a fuzzy logic controller and then sent to the PI controller for PMSM speed regulation to improve performance in both dynamic and steady-state scenarios. The

PMSM is powered by a three-phase voltage-source inverter that receives electricity from a solar photovoltaic (PV) array. Power output is maximized via an incremental conductance technique in an intermediate dc–dc converter. Response time is accelerated by a PV feed-forward term. MATLAB/Simulink is used to model and simulate the system, and a dSPACE DS-1004 controller on a prototype is used for experimental validation [34].

2.2 RESEARCH GAP

Research gaps in "Modeling and Simulation of Grid-Connected Solar PV Array System with Integrated Battery Energy Storage System" refer to the unexplored or underdeveloped areas that current studies have not adequately addressed. These include limitations in modeling real-time battery degradation, insufficient representation of environmental dynamics, lack of advanced control strategies, and minimal consideration of power quality and economic optimization. Identifying these gaps provides a foundation for developing improved, realistic, and efficient models that can contribute to more reliable, cost-effective, and sustainable solar PV-battery grid integration systems in future research.

1. Inadequate modeling of battery aging and degradation

- Most existing studies focus on ideal battery models and overlook long-term performance impacts due to degradation or cycle aging.

2. Limited consideration of real-time control strategies

- Conventional models often assume fixed control strategies without incorporating adaptive or real-time decision-making based on grid and load conditions.

3. Simplified PV system representation under dynamic environmental conditions

- Many models use constant irradiance and temperature, ignoring real-world rapid fluctuations and their effects on system stability and efficiency.

4. Lack of integration with advanced forecasting algorithms

- Most models do not incorporate solar irradiance and load forecasting for predictive energy management and optimization.

5. Neglecting power quality issues

- Harmonics, voltage fluctuations, and frequency deviations caused by PV integration are often not addressed in the simulation frameworks.

6. Insufficient exploration of hybrid storage systems

- Research often focuses only on batteries, ignoring hybrid storage options (e.g., battery + supercapacitor) which can enhance performance under variable loads.

7. Unrealistic grid interaction assumptions:

- Simulations may assume ideal grid conditions, ignoring real scenarios like weak grids, blackouts, or grid disturbances.

8. Limited validation through experimental or real-time hardware testing:

- Many simulation models are not validated with hardware-in-the-loop (HIL) or experimental setups, reducing their reliability for real-world application.

9. Insufficient focus on economic optimization:

- Studies typically prioritize technical performance over cost-benefit analysis, tariff structures, or economic dispatch optimization.

10. Cybersecurity and communication delay impacts:

- With growing smart grid integration, very few models consider cyber threats or communication delays in battery-PV-grid interactions.

2.3 OBJECTIVE

The primary objective of this thesis is to develop an accurate, dynamic, and comprehensive model of a grid-connected solar photovoltaic (PV) array system integrated with a battery energy storage system (BESS). This model is intended to simulate real-time behavior and assess the performance of the hybrid system under various environmental and operational conditions. The specific objectives include:

1. Modeling of Solar PV Array:

- To mathematically model a solar PV array using appropriate electrical equivalent circuits that reflect real-time performance under varying irradiance and temperature conditions.
- To incorporate Maximum Power Point Tracking (MPPT) algorithms such as Perturb and Observe (P&O) or Incremental Conductance for optimal energy harvesting.

2. Design and Integration of Battery Energy Storage System (BESS):

- To develop a battery model capable of simulating charging and discharging cycles, state of charge (SoC), and dynamic behavior during grid support or power backup.

- To integrate the battery system with the PV array to improve system reliability and maintain a stable power supply during solar unavailability or peak demand.

3. Grid Connection and Power Management:

- To design a control strategy for bi-directional power flow between the PV-BESS system and the utility grid.
- To ensure voltage and frequency stability, power quality, and seamless energy exchange with the grid, including during faults or islanding scenarios.

4. Simulation and Analysis:

- To simulate the entire system using platforms such as MATLAB/Simulink, analyzing performance metrics such as energy efficiency, response time, SoC levels and power quality.
- To evaluate the system's performance under various operational scenarios including partial shading, battery degradation and fluctuating load demands.

2.4 CONSTRAINTS

- **Intermittent Solar Irradiance:** The variability of solar energy due to weather changes affects consistent power generation, making accurate modeling of irradiance patterns essential.
- **Battery Degradation and Efficiency:** Battery performance declines over time due to aging, cycle degradation, and thermal conditions, which must be considered in the system design and simulations.
- **Grid Interaction Limitations:** Constraints such as voltage/frequency limits, synchronization, and anti-islanding protection affect the ability of the system to connect and operate reliably with the utility grid.
- **Converter and Control Limitations:** Power electronic components like inverters and DC-DC converters have efficiency, response time, and thermal limits, impacting the system's overall performance.
- **Computational Complexity and Simulation Time:** Accurate modeling of nonlinear components and dynamic behavior increases computational load, requiring trade-offs between simulation detail and run-time performance.

CHAPTER 3

MATHEMATICAL MODEL AND METHODOLOGY

3.1 MATHEMATICAL MODEL

A mathematical model is a formal representation of a real-world system using mathematical equations and logic to describe the relationships between input and output variables. It enables simulation, analysis, and prediction of system behavior, making it essential for design, optimization, and performance evaluation in engineering and scientific applications.

3.1.1 OUTPUT CURRENT OF A SOLAR PV ARRAY SYSTEM

$$I = I_{pv} - I_o \left[\exp \left(\frac{V + IR_s}{V_{therm} \alpha} \right) - 1 \right] - \left(\frac{V + IR_s}{R_{sh}} \right) \quad (1)$$

Where, I_{ph} =Photo-generated current (depends on solar irradiance G).

I_o = Reverse saturation Current

q = Charge of an electron (1.602×10^{-19} C).

V_{PV} =PV output voltage

R_s =Series resistance

R_{sh} =Shunt resistance

T =Temperature in Kelvin

K =Boltzmann's constant (1.38×10^{-23} J/K)

n =Diode ideality factor

The photo-generated current depends on irradiance and temperature

$$I_{pv} = (I_{ph_ref} + \alpha(T - T_{ref})) \frac{G}{G_{ref}} \quad (2)$$

G = Instantaneous solar irradiance

G_{ref} = Reference irradiance (1000 W/m^2)

α = Temperature coefficient of short-circuit current

T_{ref} = Reference temperature

$$I_{pv} = \left(\frac{R_{sh} + R_s}{R_{sh}} \right) I_{sc} \quad (3)$$

$$\text{The DC (PV) power output is: } P_{PV} = V_{PV} * I_{PV} \quad (4)$$

3.1.2 DIFFERENTIAL EQUATION FOR PV OUTPUT VOLTAGE

$$C_{PV} \frac{dV_{PV}}{dt} = I_{PV} - I_{load} \quad (5)$$

C_{PV} =Capacitance of the PV module

I_{load} =Current drawn by the connected load.

The battery model defining the voltage V_b in terms of State of Charge (SOC):

$$V_b = E - R_b I_b - K \frac{Q}{Q - SOC \cdot Q} I_b \quad (6)$$

E = Open-circuit voltage

R_b = Internal resistance

I_b = Charging/discharging current

K = Polarization resistance

Q = Total battery capacity

3.1.3 SOC DIFFERENTIAL EQUATION:

$$\frac{dSOC}{dt} = \frac{n}{C_{bat}} I_b \quad (7)$$

n = Battery efficiency

Battery capacity in Ah

$$V_{Batt} = V_o - Ri - \frac{Q}{Q - it} (it + t^*) + \text{Exp}(t) \quad (8)$$

$$V_{Batt} = V_o - Ri - \frac{Q}{it - 0.1Q} (i^*) - [\frac{Q}{Q - it} (it) + \text{Exp}(t)] \quad (8)$$

$$P(t) = \frac{2}{T} \int_{t-\frac{\tau}{2}}^t (V_t(\tau) i_c(\tau)) d\tau = \frac{V_t(t) V_c(t)}{\omega L_c} \sin \alpha \quad (9)$$

$$P(t) = \frac{V_t(t) V_c(t)}{\omega L_c} \alpha \quad (10)$$

$$Q(t) = \frac{V_t(t)}{\omega L_c} (V_t(t) - V_c(t)) \quad (11)$$

$$\delta^* = K_{p1} * (P_{MPPref} - P_{PV}) + K_{i1} * \int_0^t (P_{MPPref} - P_{PV}) dt \quad (12)$$

$$S^* = K_{p5} (P_{Batterref} - P_{Batter}) + K_{i5} * \int_0^t (P_{Batterref} - P_{Batter}) dt \quad (13)$$

$$V_{c1}^*(t) = v_1(t) [1 + K_{p2} (V_{(t)}^* - V_{(t)})] + K_{i2} \int_0^t (V_{(t)}^*(t) - V_{(t)}(t)) dt \quad (14)$$

$$\alpha_{1}^* = K_{p3} (f_{ref} - f_{measured}) + K_{i3} * \int_0^t (f_{ref} - f_{measured}) dt \quad (15)$$

$$\alpha_{2}^* = K_{p4} (1.02 * P_{AC} - P_{DC}) + K_{i4} * \int_0^t (1.02 * P_{AC} - P_{DC}) dt \quad (16)$$

$$\alpha^* = \frac{\alpha_{1}^* + \alpha_{2}^*}{2} \quad (17)$$

$$V_{c1}^* = K_{p2} (Q_{ref} - Q_{actual}) + K_{i2} * \int_0^t (Q_{ref} - Q_{actual}) dt + 1) V_{(t)} \quad (18)$$

$$\alpha_{1}^* = K_{p3} (P_{ref} - P_{actual}) + K_{i3} * \int_0^t (P_{ref} - P_{actual}) dt \quad (19)$$

$$\alpha_{2}^* = K_{p4} (1.02 * P_{ACmeasured} - P_{DC}) + K_{i4} * \int_0^t (1.02 * P_{ACmeasured} - P_{DC}) dt \quad (20)$$

$$\alpha^* = \frac{\alpha_1^* + \alpha_2^*}{2} \quad (21)$$

Where,

P_{PV}	PV active Power output
α_1^*	Phase shift
$P_{Battref}$	Reference Battery active power
δ^*	Duty Cycle of DC/Dc Boost Converter.
P_{Batt}	actual injected power
f_{ref}	Reference micogrid frequency
$f_{measured}$	Measured micogrid frequency
P_{AC}	AC side total average active power
P_{DC}	DC side average active power
Q_{ref}	Average reactive power reference (Reactive load)
Q_{actual}	Actual generated reactive power
P_{actual}	Actual generated active power
Q_{ref}	Average active power reference (Active load)

3.2 METHODOLOGY

The Maximum Power Point Tracking (MPPT) algorithm is a technique used in solar photovoltaic (PV) systems to extract the maximum possible power from a PV array under varying environmental conditions such as irradiance and temperature. Due to the non-linear I-V (current-voltage) characteristics of solar panels, there exists a unique operating point where the product of current and voltage (i.e., power) is maximum—this is called the Maximum Power Point (MPP). The MPPT algorithm continuously adjusts the operating voltage or current of the PV panel to locate and maintain this point, ensuring optimal energy conversion.

Several MPPT techniques are commonly used, including Perturb and Observe (P&O), Incremental Conductance, and Fuzzy Logic Control. These algorithms vary in complexity, speed, and accuracy. MPPT is essential for improving the efficiency of PV systems, especially in conditions where sunlight and temperature fluctuate. It is typically implemented in the control software of a DC-DC converter or inverter within the solar power system.

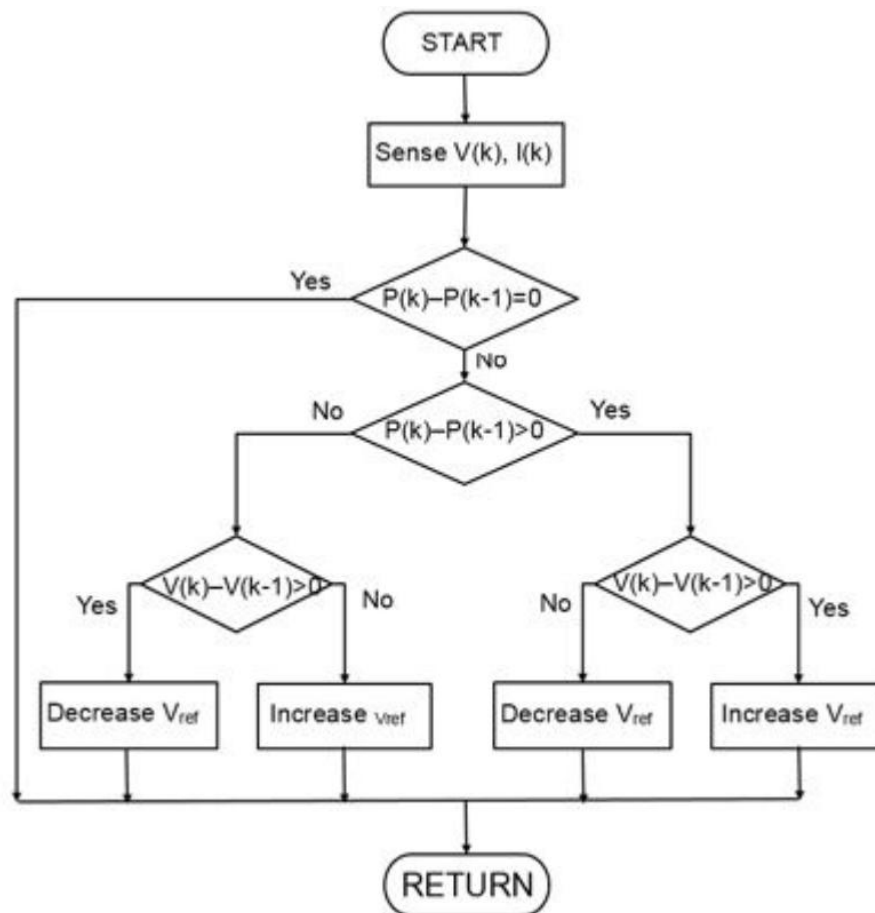


Figure 3.1: MPPT loop diagram.

In Figure 3.1 MPPT Algorithm shows that how we maximize output power.

CHAPTER 4

RESULTS AND DISCUSSIONS

4.1 OUTPUT OF SOLAR PV ARRAY SYSTEM

It was observed that the SOLTON Power SPI-215P modules, configured with 10 modules connected in series and 47 such strings connected in parallel, exhibited varying electrical performance under different operating conditions. Specifically, the generated direct current (DC) and DC power output fluctuated in response to changes in voltage levels and time intervals, which are inherently influenced by environmental factors such as solar irradiance and temperature. The series connection of 10 modules effectively increased the system's overall voltage, while the parallel configuration of 47 strings enhanced the current-carrying capacity of the array. These variations in output highlight the dynamic nature of photovoltaic (PV) systems, where real-time conditions significantly impact power generation. Such behavior underscores the importance of accurate modeling and real-time monitoring in PV system design and simulation, particularly when integrated with energy storage and grid interfaces, to ensure optimal energy harvesting and efficient power management throughout the system's operation.

Array type: SOLTON Power SPI-215P;
10 series modules; 47 parallel strings

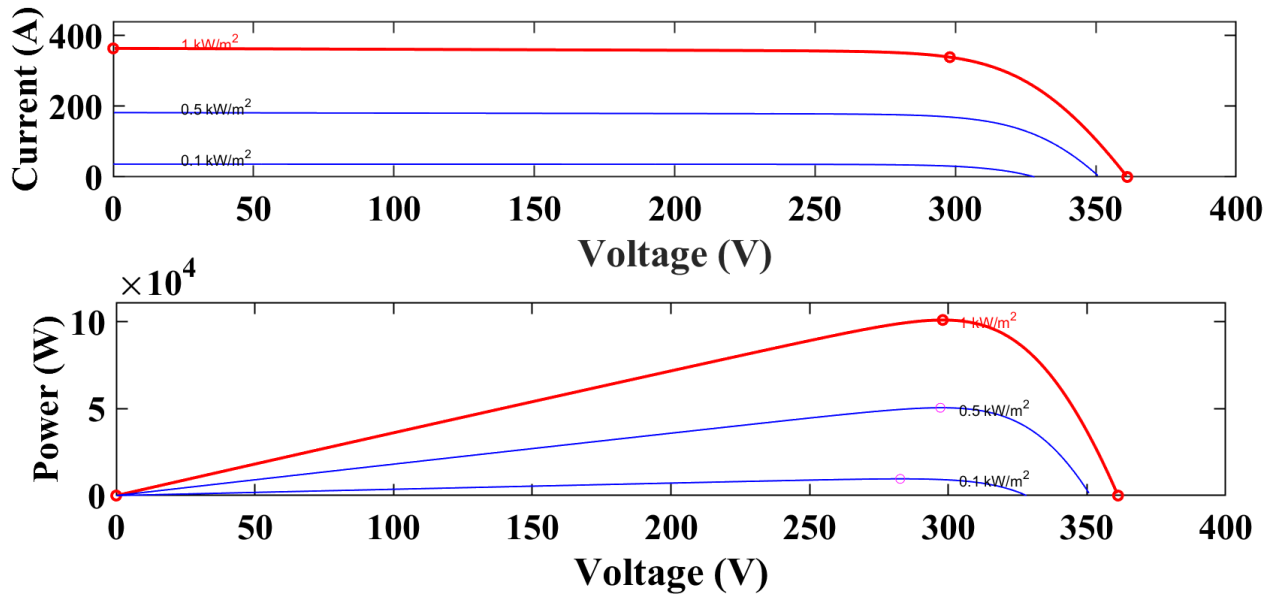


Figure 4.1: Shows the output current and output power at different voltages.

The data presented in the table captures a comprehensive set of parameters obtained from the solar panel operating under standard test conditions—solar irradiance of 1000 W/m² and a cell temperature of 25°C—measured across different time intervals throughout the day. These parameters include open-circuit voltage (V_{oc}), short-circuit current (I_{sc}), maximum power point (P_{max}), fill factor (FF), and efficiency (η), among others. The values exhibit temporal variation, which is directly influenced by changing solar irradiance and ambient conditions. This variation highlights how solar panel performance is dynamic and dependent on real-time environmental factors. The detailed observation of these parameters over time is crucial for analyzing the PV system's energy output, optimizing system design, and improving forecasting accuracy. Furthermore, this tabular dataset provides a valuable reference for validating simulation models and assessing system behavior under fluctuating solar resource availability. It emphasizes the importance of continuous monitoring to ensure optimal operation and energy yield from solar PV installations.

Table 4.1: PV Panel Parameters at 1000W/m² and 25 °C.

Model	Data
P_{DC}	600W
V_{MPPT}	15.35V
I_{MPPT}	7.61A
V_{oc}	32.90V
I_{sc}	8.21A

4.2 BATTERY MODEL

The Simulink model of the battery system demonstrates the dynamic behavior of cell balancing and charging in response to the varying DC power generated by the solar PV array over the course of a typical day. The model effectively simulates how fast cell balancing mechanisms are activated to equalize the state of charge (SoC) among individual battery cells, thereby preventing overcharging or undercharging of specific cells and ensuring uniform performance. As the solar irradiance fluctuates throughout the day, resulting in corresponding variations in the PV array's DC output, the battery charging process adjusts accordingly. The model reflects real-time adaptation of charging currents, maintaining battery health and extending its lifecycle. This simulation highlights the critical role of intelligent battery management systems (BMS)

in maintaining operational efficiency and reliability. The results reinforce the importance of synchronized energy storage integration for maximizing the effectiveness of solar PV systems, particularly in grid-connected configurations with fluctuating energy inputs.

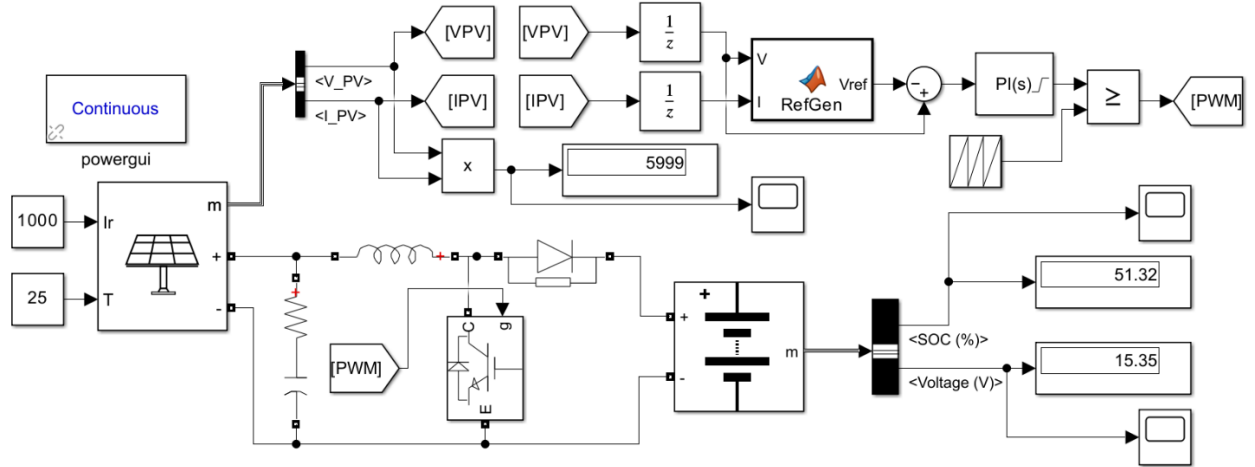


Figure 4.2: Simulink Model of a battery.

4.3 SOC AND VOLTAGE OF BATTERY

4.3.1 SOC OF BATTERY

The graph clearly illustrates that the battery initiates charging at time $t=10$ seconds, indicating the onset of energy transfer from the solar PV system to the energy storage unit. This behavior reflects the dynamic interaction between the generated DC power from the solar array and the battery's charging mechanism. As the DC source—derived from the solar PV system—varies throughout the day due to changing irradiance and environmental conditions, the battery management system (BMS) responds by regulating the charging current and initiating fast cell balancing processes. Cell balancing ensures that all individual cells within the battery pack maintain uniform voltage and state of charge (SoC), preventing imbalances that could lead to performance degradation or safety concerns. This simulation emphasizes how direct DC coupling enables efficient energy transfer and adaptive battery control, ensuring optimal storage utilization. The graphical result serves as a practical validation of the model's responsiveness to real-time solar power generation.

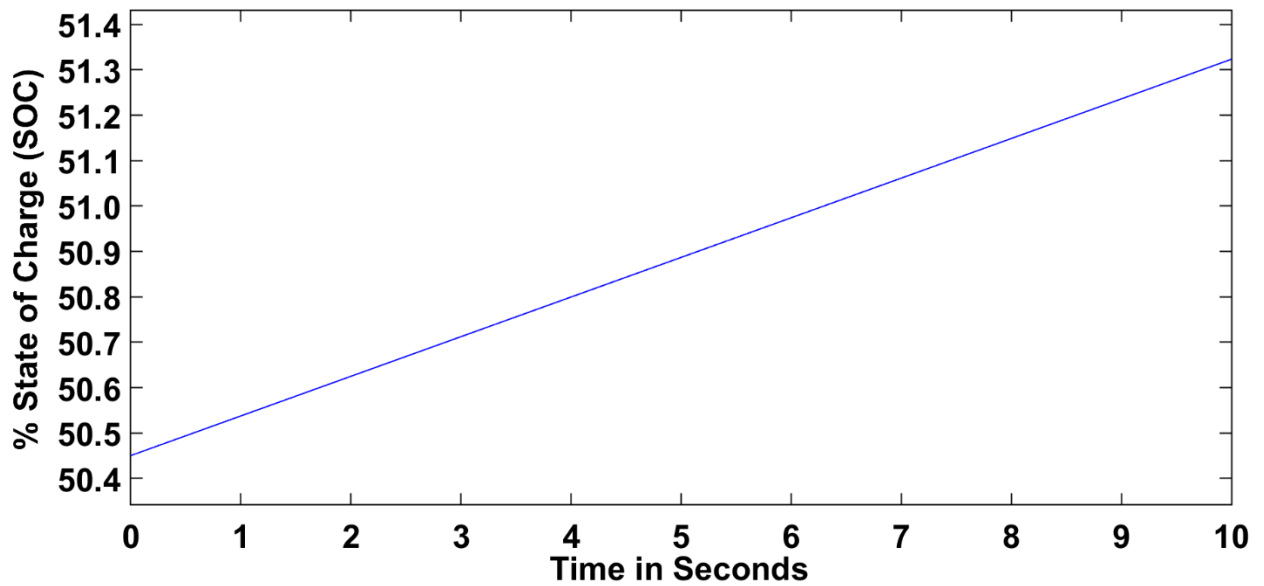


Figure 4.3.1: Charging of a Battery for 10 sec.

4.3.2 VOLTAGE OF BATTERY

The graph clearly indicates that the battery begins its charging process at **t = 10 seconds**, demonstrating a prompt and controlled response of the Battery Energy Storage System (BESS) to the direct supply of DC power generated by the solar photovoltaic (PV) array. This voltage-versus-time graphical representation offers critical insight into the dynamic interaction between solar generation and energy storage. As the battery starts charging, a noticeable increase in battery terminal voltage is observed, confirming that energy is being transferred from the PV source to the storage system.

The charging profile shown in the graph is not static; rather, it evolves in real time, influenced by fluctuating solar irradiance levels throughout the day. These variations are caused by environmental factors such as the sun's angle, cloud movement, and atmospheric clarity. As a result, the rate at which the battery charges including the slope of voltage rise is directly affected by the instantaneous availability of solar energy. This responsiveness is crucial in renewable energy systems where power generation is inherently intermittent.

The graph also underscores the efficiency of the battery management system (BMS) in regulating the charging process. The smooth and gradual voltage increase indicates that the BMS is effectively managing the cell balancing and thermal conditions to avoid overcharging, voltage mismatch, or potential degradation. This ensures the longevity and safety of the battery while optimizing the storage of excess solar energy. In essence, the observed charging behavior validates the reliability and practicality of integrating a

BESS with a solar PV system. It demonstrates the system's ability to capture and store solar energy during periods of generation and subsequently make it available during demand peaks or when solar power is insufficient. This capability is fundamental to achieving grid stability, energy self-sufficiency, and efficient utilization of renewable resources. The analysis further reinforces the importance of real-time monitoring and simulation in optimizing the performance of solar-battery hybrid systems, making it a critical component in modern energy infrastructure design and planning.

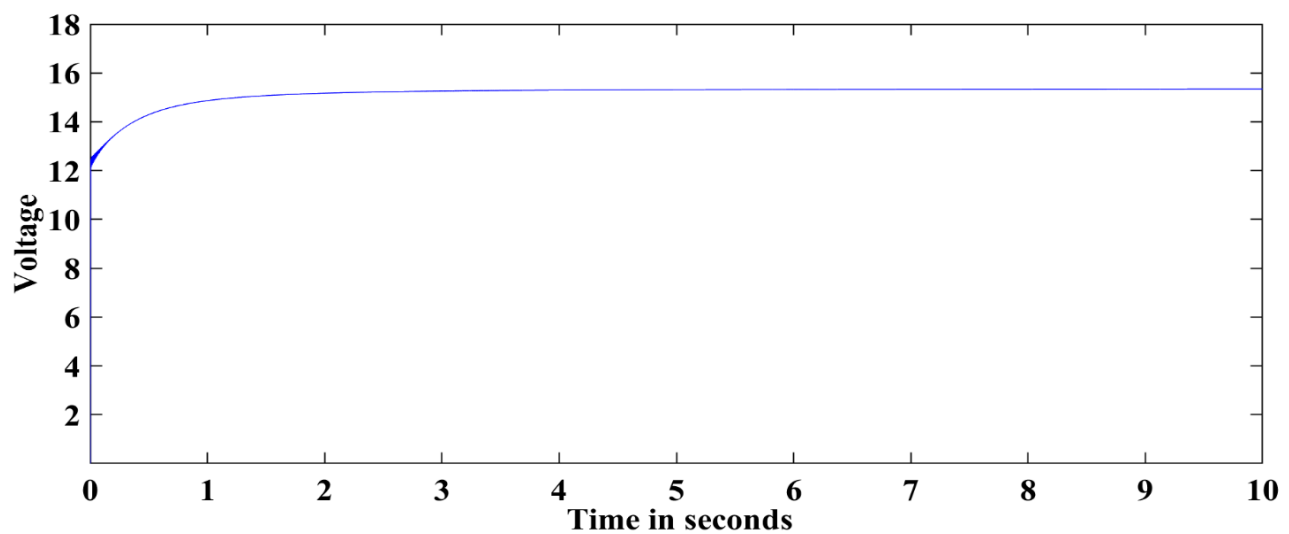


Figure 4.3.2: Voltage across Battery after charging for 10 sec.

4.4 DC RESPONSE

The graph clearly illustrates the variation in DC power output generated by the solar panel throughout the day under standard test conditions, with solar irradiance maintained at 1000 W/m^2 and ambient temperature at 25°C . These conditions represent the optimal environment for photovoltaic (PV) operation, allowing the system to achieve peak performance. The graph highlights how the solar irradiance profile, influenced by the sun's position and atmospheric conditions, directly affects the power generation at different time intervals. During periods of high solar intensity typically around midday the graph shows a corresponding peak in DC power output, while lower values are observed during early morning and late afternoon hours. This behavior demonstrates the dependency of solar power generation on real-time environmental conditions and underlines the importance of accurate modeling for predicting energy yield. The graphical analysis serves as a crucial tool in evaluating system performance,

sizing components and designing efficient energy storage and grid integration strategies.

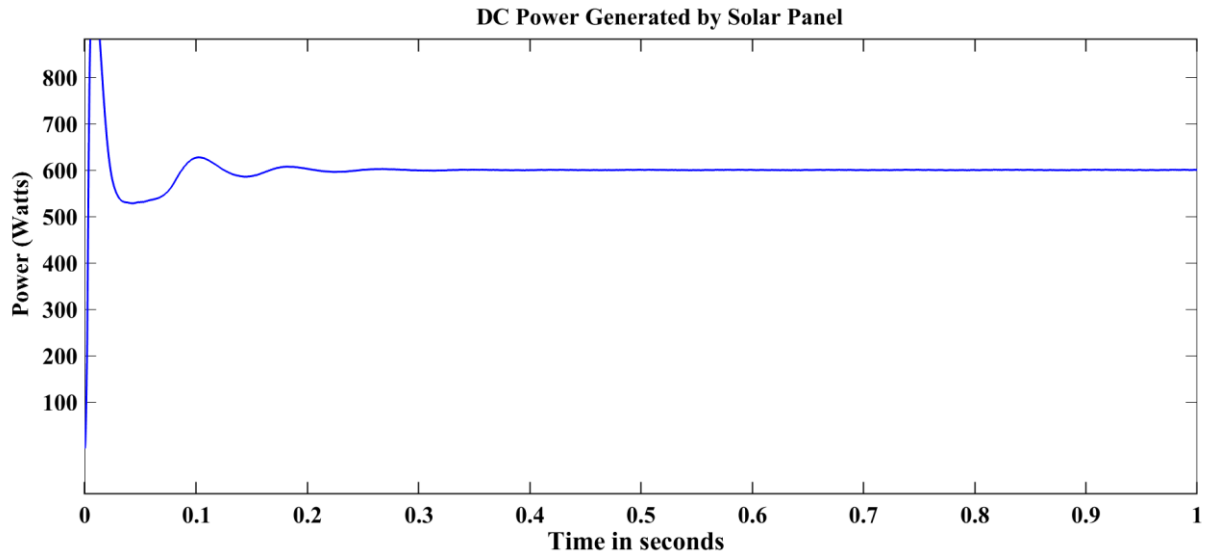


Figure 4.4: DC Power generated by Solar Panel.

4.5 COMPLETE SIMULINK MODEL

The Simulink model effectively demonstrates the comprehensive process of energy conversion and power transfer in a grid-connected solar photovoltaic (PV) system integrated with a Battery Energy Storage System (BESS). It begins with the solar PV array capturing solar irradiance and converting it into electrical energy in the form of direct current (DC). This generated DC power is first utilized to charge the battery, allowing energy storage for periods of low solar generation or peak load demand. The charging behavior of the BESS dynamically varies based on real-time solar irradiance, ambient conditions, and the energy consumption profile.

The model further simulates the operation of a DC-AC inverter, which converts the remaining or stored DC power into alternating current (AC) suitable for grid synchronization and supply. This AC power is either exported to the utility grid or used to fulfill local load requirements, depending on energy availability and demand. Through this simulation, the model highlights the system's ability to adapt to fluctuating environmental conditions while ensuring reliable and efficient energy flow.

Overall, the Simulink model offers critical insights into component interaction, system timing, and energy conversion efficiency, confirming the operational viability and reliability of hybrid solar-BESS configurations in modern grid-tied energy systems.

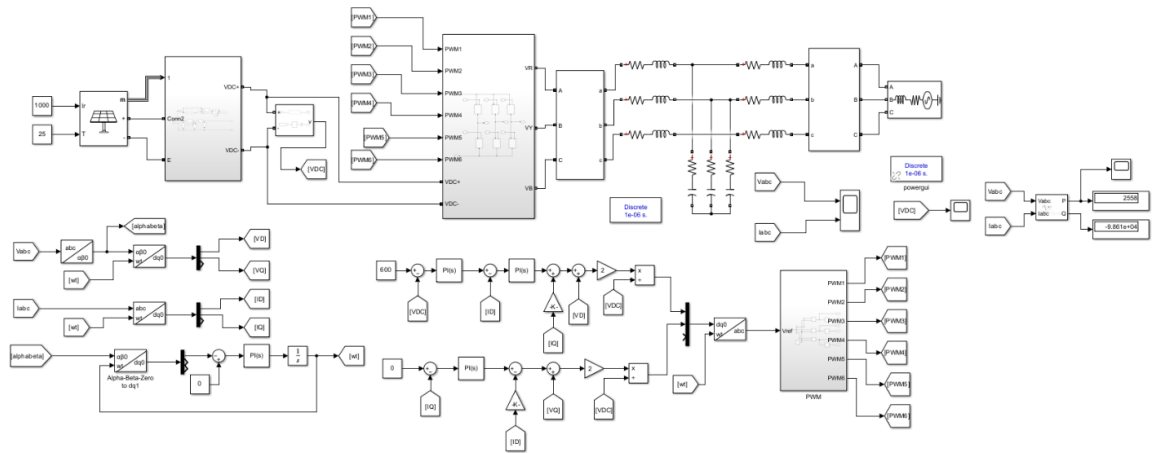


Figure 4.5: Simulink Modeling and Simulation of Grid Connected Solar PV Array System with Integrated Battery Energy Storage system.

The data presented in the given table clearly demonstrates that the amount of energy generated by the solar photovoltaic (PV) system is significantly influenced by two primary environmental factors: solar irradiance and cell temperature, observed at various time intervals throughout the day. Solar irradiance, measured in W/m^2 , directly affects the intensity of sunlight incident on the PV modules, which governs the level of electrical power output. Higher irradiance levels correspond to increased photon availability, thus enhancing the current and power generation capacity of the solar cells. Conversely, cell temperature plays a critical role in determining the voltage output; as temperature increases, the voltage typically decreases due to the negative temperature coefficient of the semiconductor material used in PV cells.

Table 4.2: Irradiance V/S Cell Temperature ($^{\circ}\text{C}$).

Scenario	Irradiance (W/m^2)	Cell Temperature ($^{\circ}\text{C}$)	Description
Standard Test Condition (STC)	1000	25	Ideal lab testing condition. Baseline for PV performance.
High irradiance, hot day	1000	45	Typical of sunny summer afternoons lower efficiency due to heat.
Medium irradiance, mild weather	600	30	Cloudy or hazy condition moderate power output.

4.6 AC RESPONSE

The graph clearly illustrates the successful conversion of DC power, generated by the solar photovoltaic (PV) array, into three-phase AC power, making it suitable for grid integration. This conversion process is achieved through a DC–AC inverter, which plays a critical role in interfacing renewable energy systems with the utility grid. As shown in the graph, the three-phase AC output closely follows the fluctuations of the input DC power at different time intervals throughout the day, reflecting the dynamic nature of solar energy generation driven by irradiance and temperature variations. The waveform characteristics confirm that the inverter maintains voltage and frequency stability, ensuring synchronization with grid parameters. This consistent performance across the day underscores the system's ability to adapt to real-time solar conditions while meeting grid standards. The graphical representation thus validates the operational reliability and effectiveness of the inverter in facilitating seamless power exchange between the PV system and the electrical grid.

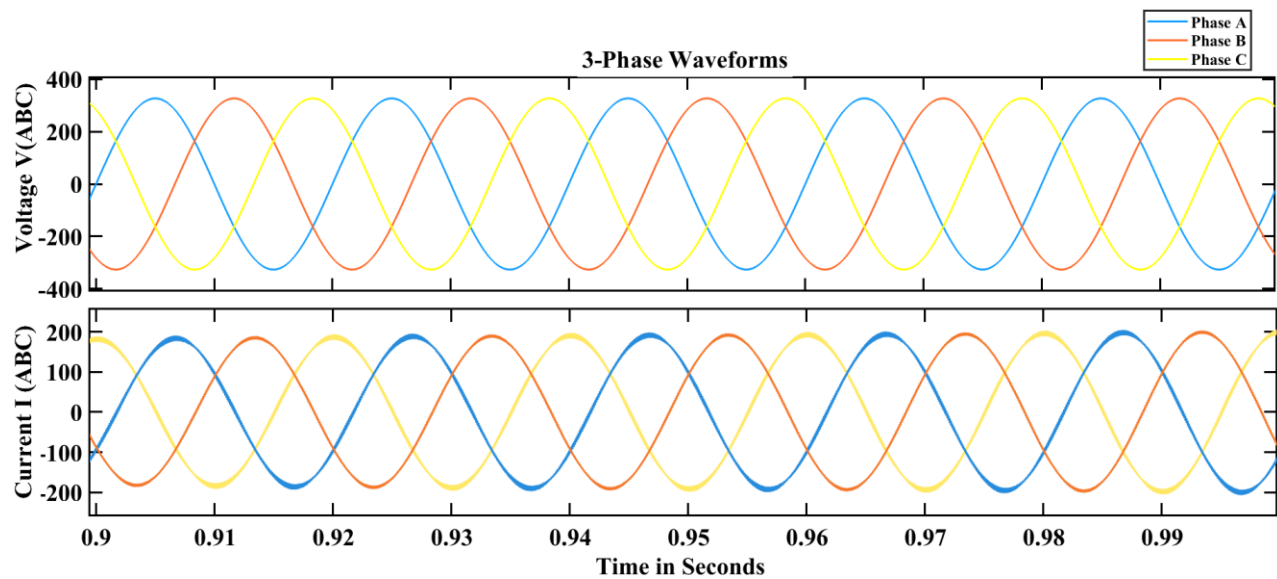


Figure 4.6: AC response of 3-phase grid.

CHAPTER 5

SUMMARY AND CONCLUSIONS

5.1 SUMMARY

This thesis presents the detailed modeling and simulation of a grid-connected solar photovoltaic (PV) array system integrated with a Battery Energy Storage System (BESS), aimed at enhancing the efficiency, reliability and stability of renewable energy supply. The PV array serves as the primary source of clean energy, while the BESS ensures continuous power availability by storing excess energy during high generation periods and supplying power during low irradiance or peak demand conditions. A DC-DC boost converter with Maximum Power Point Tracking (MPPT) is implemented to optimize PV output and a bidirectional converter manages battery charging and discharging. The DC power is then converted to grid-compatible AC via a DC-AC inverter. Grid interface modeling includes synchronization, voltage regulation and power flow control. Simulations are conducted under varying environmental and load conditions to assess system performance. Results demonstrate improved energy management, reduced grid dependency and enhanced integration of intermittent solar energy into the power system.

5.2 CONCLUSION

In conclusion, the modeling and simulation of a grid-connected solar photovoltaic (PV) array system integrated with a Battery Energy Storage System (BESS) successfully demonstrate the potential for enhanced energy reliability, efficiency and grid stability. The simulation framework developed in this thesis effectively captures the dynamic behavior of each system component, including the PV array, MPPT-controlled DC-DC boost converter, bidirectional battery converter, DC-AC inverter, and grid interface. Results indicate that integrating BESS enables optimal utilization of solar energy by mitigating the effects of intermittency, enabling peak load management and ensuring power availability during periods of low solar irradiance or grid disturbances. Furthermore, the system provides ancillary grid services such as frequency and voltage regulation, contributing to overall grid resilience. The study validates that such integrated systems are crucial for advancing the adoption of renewable energy in modern power networks and provides a strong foundation for future developments in smart grid and energy management strategies.

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